Llm Rag Notes

Transformer

The Transformer is a neural network architecture designed for sequence processing. It relies heavily on the attention mechanism rather than recurrence. Transformers are widely used in modern language models.

Attention Mechanism

The attention mechanism helps a model focus on the most relevant parts of the input when producing an output. Self-attention allows each token to consider relationships with other tokens in the same sequence.

Large Language Models (LLMs)

Large Language Models are AI models trained on very large amounts of text data. They can understand and generate human-like language. Common tasks include question answering, summarization, translation, coding and conversation.

Generative AI

Generative AI refers to AI systems that can create new content such as text, images, audio, video and code. LLMs are a major type of generative AI used for text generation.

Prompt Engineering

Prompt engineering is the process of designing clear and effective instructions for an AI model. A good prompt provides context, constraints and the desired output format.

Embeddings

Embeddings are numerical vector representations of text, images or other data. Similar meanings are represented by vectors that are close together in vector space. Embeddings are commonly used for semantic search and RAG systems.

Vector Database

A vector database stores embeddings and performs similarity searches. Instead of searching only for exact keywords, it can retrieve information based on semantic similarity. Examples include Pinecone, Chroma and Weaviate.

Pinecone

Pinecone is a managed vector database used to store, index and search vector embeddings. It supports similarity search and can organize vectors using namespaces. Pinecone is commonly used in Retrieval-Augmented Generation applications.

RAG - Retrieval-Augmented Generation

Retrieval-Augmented Generation combines information retrieval with a Large Language Model. A user question is converted into an embedding, relevant information is retrieved from a knowledge base, and the retrieved context is given to the LLM to generate a grounded answer.

Multi-RAG

Multi-RAG uses multiple knowledge sources, retrievers or namespaces. A query router can identify the topic of a user question and send it to the most relevant knowledge base. For example, an AI and Machine Learning namespace can be separate from an LLM and RAG namespace.

Basic RAG Pipeline

Documents are collected and split into smaller chunks. Each chunk is converted into an embedding and stored in a vector database. When a user asks a question, the system creates a query embedding, retrieves similar chunks and sends them with the question to an LLM.

Hallucination in LLMs

Hallucination occurs when an AI model generates information that is incorrect or unsupported. RAG can reduce hallucination by providing relevant retrieved context, although retrieval quality and prompt design remain important.